

Atanu Saha

Backend Engineer | Event-Driven Microservices | Claude Certified Architect

Dhaka, Bangladesh • Open to Relocation • +880-1828675754 • atanu.saha415@gmail.com
linkedin.com/in/atanusaha143 • github.com/Atanusaha143 • atanusaha143.github.io/portfolio

PROFESSIONAL SUMMARY

Backend engineer and ICPC Asia West Continent Finalist with 3+ years building event-driven async Python microservices for checkout and payments (FastAPI, NATS, PostgreSQL, Stripe, Adyen) on GCP and AWS, serving ~87K requests/day.

WORK EXPERIENCE

Cefalo Bangladesh Ltd. | Client: [TicketCo AS](#) (Norway)

Dhaka, Bangladesh

Software Engineer

Jan 2026 – Present

- Merged a background worker and its cron scheduler into one FastAPI app (FastStream consumers, APScheduler jobs, one event loop): **~38% lower peak memory**, ~20% lower peak CPU, no stuck messages.
- Resolved a missing-index incident via SQLAlchemy models and idempotent Alembic migrations, rolled out with **zero downtime across 2 services**; then cut N+1 queries and set load-test baselines.
- Implemented Stripe and Adyen payment, refund, and split-payment (Adyen for Platforms) APIs with HMAC-validated webhooks routed to a dedicated handler per payment model.
- Built PyTest suites from scratch across services, reaching **2000+ tests** gated in CI on every PR; migrated 2 services to Poetry with CI lock checks and grouped Dependabot updates.
- Structured 2 FastAPI codebases for AI coding agents (Claude Code skills, subagents, enforced conventions).

Associate Software Engineer

Oct 2023 – Dec 2025

- Joined a new checkout product at inception (Jun 2024) and architected 3 event-driven microservices (FastAPI, FastStream, NATS), sustaining **~87K requests/day at ~62ms** average response time.
- Cut NATS publish-time spikes from **0.13–254s to 30–40ms** via connection pooling and tuning.
- Refactored the refund API to async with retries, lowering p95 response time by **~72%** (~2.5s to ~700ms).
- Hardened failure handling with timeouts, retries, and circuit breakers; added structured logging and correlation IDs; reached **85%+ coverage** (PyTest, Cucumber BDD) under CI gates.

EDUCATION & CERTIFICATIONS

B.Sc. in Computer Science and Engineering | American International University-Bangladesh

CGPA 3.98/4.00; Summa Cum Laude (Gold Medal)

May 2018 – Dec 2021

Dr. Anwarul Abedin Leadership Award (22nd Convocation, Feb 2025); Dean's List; fully funded scholarship

- **Claude Certified Architect – Foundations**, Anthropic, Sep 2026 ([Credly](#))

TECHNICAL SKILLS

- **Languages & APIs:** Python, SQL, TypeScript, asyncio, REST APIs
- **Frameworks:** FastAPI, FastStream, SQLAlchemy, Alembic, Pydantic, APScheduler
- **Data & Messaging:** PostgreSQL (indexing, query tuning, migrations), NATS JetStream, Valkey
- **Payments & Cloud:** Stripe, Adyen, GCP (Cloud Run, Cloud SQL), AWS (ECS, ECR), Docker
- **DevOps & Testing:** GitHub Actions CI/CD, Poetry, PyTest, Cucumber BDD, New Relic
- **AI Tooling:** Claude Code (skills, subagents, hooks), Claude Agent SDK, Claude API, MCP, RAG

ACHIEVEMENTS & LEADERSHIP

- **Claude Workshop Co-author & Co-facilitator**, Cefalo (Aug 2026 – Present): sessions for 30+ engineers
- **Competitive Programming Trainer**, AIUB (Sep 2022 – Mar 2024): trained 100+ students in algorithms
- **ICPC:** Honorable Mention, 2022 Asia West Continent Final; 11th place, 2022 Asia Dhaka Regional ([profile](#))
- **Publication:** “A Comparative Analysis on Fake News Detection Methods,” ICCA '22 ([ACM](#)), Mar 2022